

# Supplementary Material

## Low-Density lipoprotein triglyceride and coronary heart disease risk: evidence from Mendelian Randomization

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Short running title: LDL-TG and CHD Risk : Mendelian Randomization Evidence

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## Materials and Methods

### LDL-TG specific GWAS on UKB participants

Individuals with valid LDL-TG nuclear magnetic resonance (NMR) measurement were included. At UKB, NMR biomarkers were measured from baseline EDTA plasma samples obtained from non-fasting blood draws in a random subset of UK Biobank participants [1]. We excluded participants who had withdrawn consent, had discordance between genetically inferred and self-reported sex, or had sex-chromosome aneuploid (Supplementary Figure 1). To reduce bias from population stratification, we further restricted the analysis to participants with genetically inferred European ancestry using a K-means clustering algorithm with  $K = 4$ , excluding those classified as non-European ancestry [2]. Before association testing, LDL-TG values were inverse rank-normalized to have a mean of 0 and standard deviation of 1, enabling scale-free interpretation of genetic effect estimates [3]. GWAS was performed using BOLT-LMM, a linear mixed model approach that accounts for cryptic relatedness and population structure through the random-effect components [4]. Age, sex, ApoB, fasting time, defined as the interval between last caloric intake and blood draw, and genotyping array, defined as a binary indicator for the UK BiLEVE array versus UKB Axiom array, were adjusted in the LDL-TG GWAS model. Because BOLT-LMM accounts for relatedness and population structure in mixed-model framework, genetic principal components were not included as fixed-effect covariates [3]. We used PLINK2 [5] to perform genotype quality control. According to the reference [4], the PLINK2 parameters were defined as follow: variants were retained if they had a minor allele frequency (MAF)  $\geq 0.01$  and minor allele count (MAC)  $\geq 20$  to ensure sufficient information content were captured at lower-frequency sites [6, 7]. The single nucleotide polymorphism (SNP)s with genotype missingness  $> 0.01$  and individuals with genotype call rate  $\leq 0.9$  were excluded [8].

### Genetic instrument selection

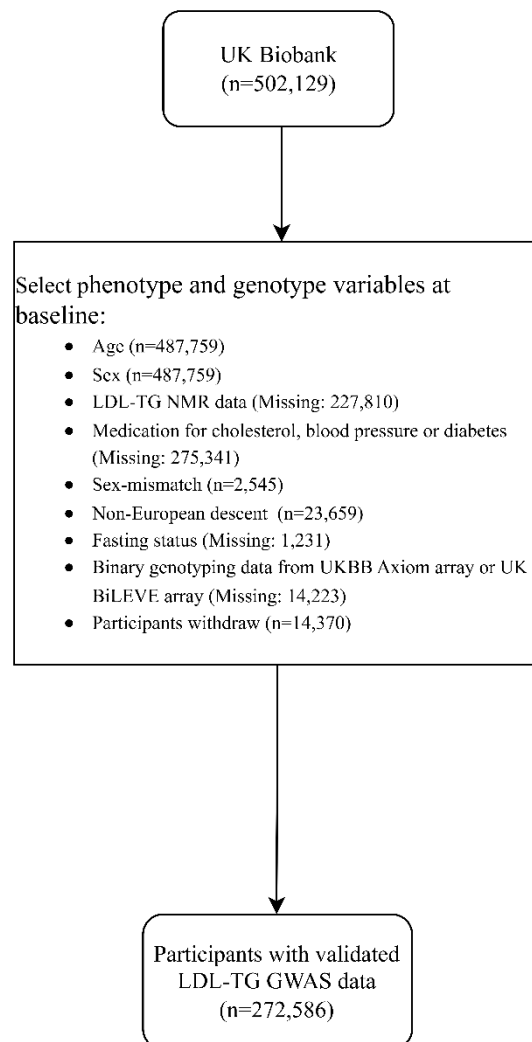
To construct genetic instruments for LDL-TG, the exposure variable, we first selected SNPs associated with LDL-TG at genome-wide significance ( $P < 5 \times 10^{-8}$ ) [9]. We additionally constructed a relaxed instrument set using a suggestive association threshold of  $P < 1 \times 10^{-6}$ , with weak-instrument assessment applied throughout [10]. To ensure independence among instruments, linkage disequilibrium (LD) clumping was performed using European ancestry reference panel from the 1000 Genomes Project Consortium [11]. Independent variants were retained using  $r^2 < 0.001$  within a 10,000-kb window, keeping the most strongly associated SNP within each locus [12]. For each retained SNP, we calculated the first stage F-statistic and excluded instruments with  $F < 10$  to reduce potential weak instrument bias [13].

The summary statistics of CHD, the outcome, were obtained from the GWAS Catalog database: <https://www.ebi.ac.uk/gwas/home>, with study accession ID: GCST90132314. In total, this study included 1,165,690 participants, comprising 181,522 CHD cases and 984,168 controls [14].

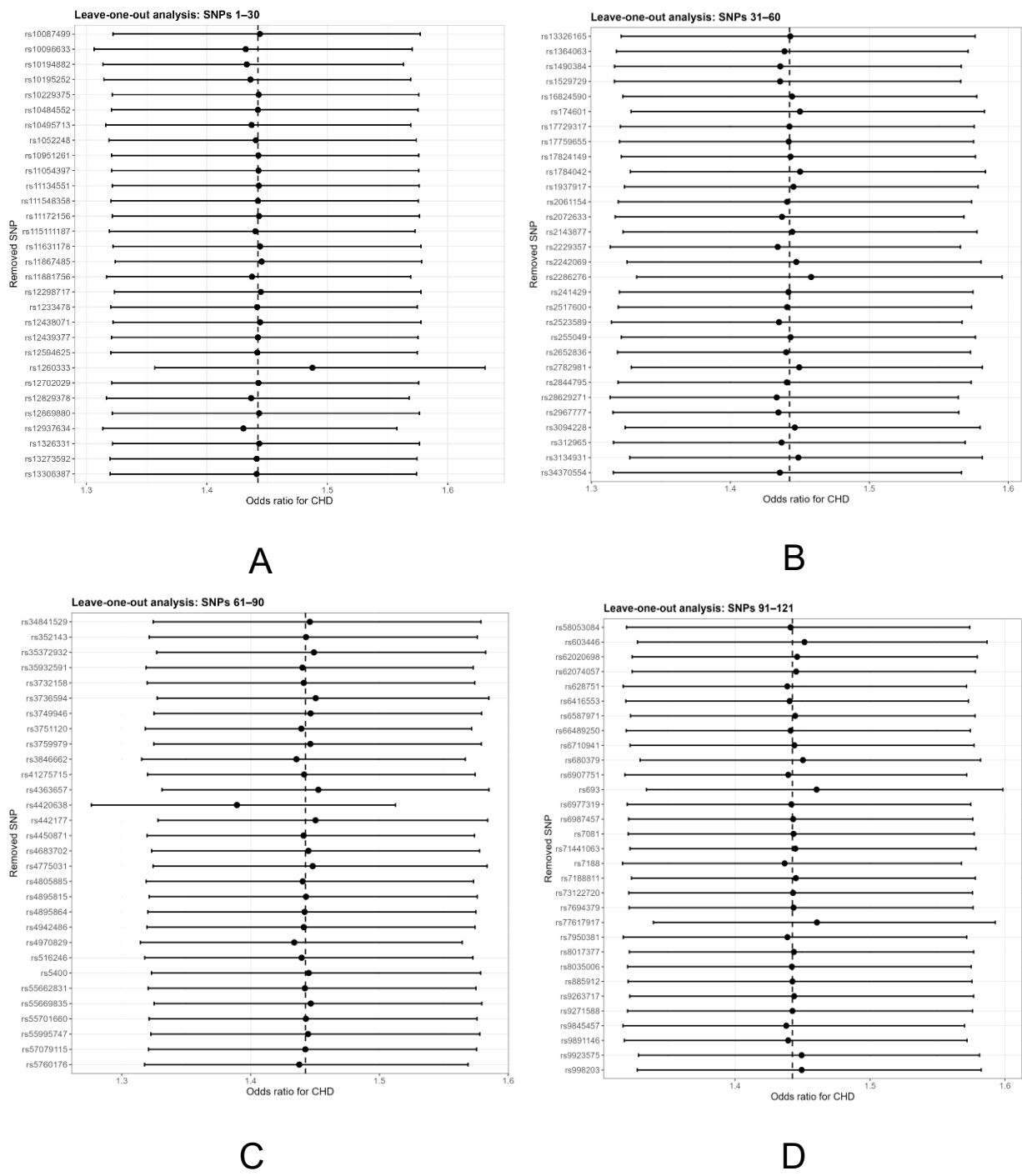
### Mendelian randomized analyses

Exposure and outcome summary statistics were harmonized by aligning effect alleles across the LDL-TG and CHD GWAS datasets. Palindromic A/T and C/G variants were evaluated using effect-allele frequencies to resolve potential strand ambiguity. Variants with unresolved ambiguity, particularly those with  $MAF \approx 0.5$ , were excluded [15]. We used the Inverse-variance weighted (IVW) method as the primary two-sample MR approach to estimate the causal effect of genetically predicted LDL-TG on CHD risk [16]. Steiger filtering was applied to confirm that genetic instruments explained more variance in LDL-TG than in CHD [17]. To evaluate the robustness of the MR findings and assess potential horizontal pleiotropy, we performed MR-Egger regression and weighted-median analysis [12, 18]. In addition, leave-one-out analyses were conducted to assess whether the overall MR estimate was driven by any single instrument. For loci with complex LD or prior pleiotropy concerns, we additionally performed colocalization to verify that the exposure and outcome share the same causal signal [19].

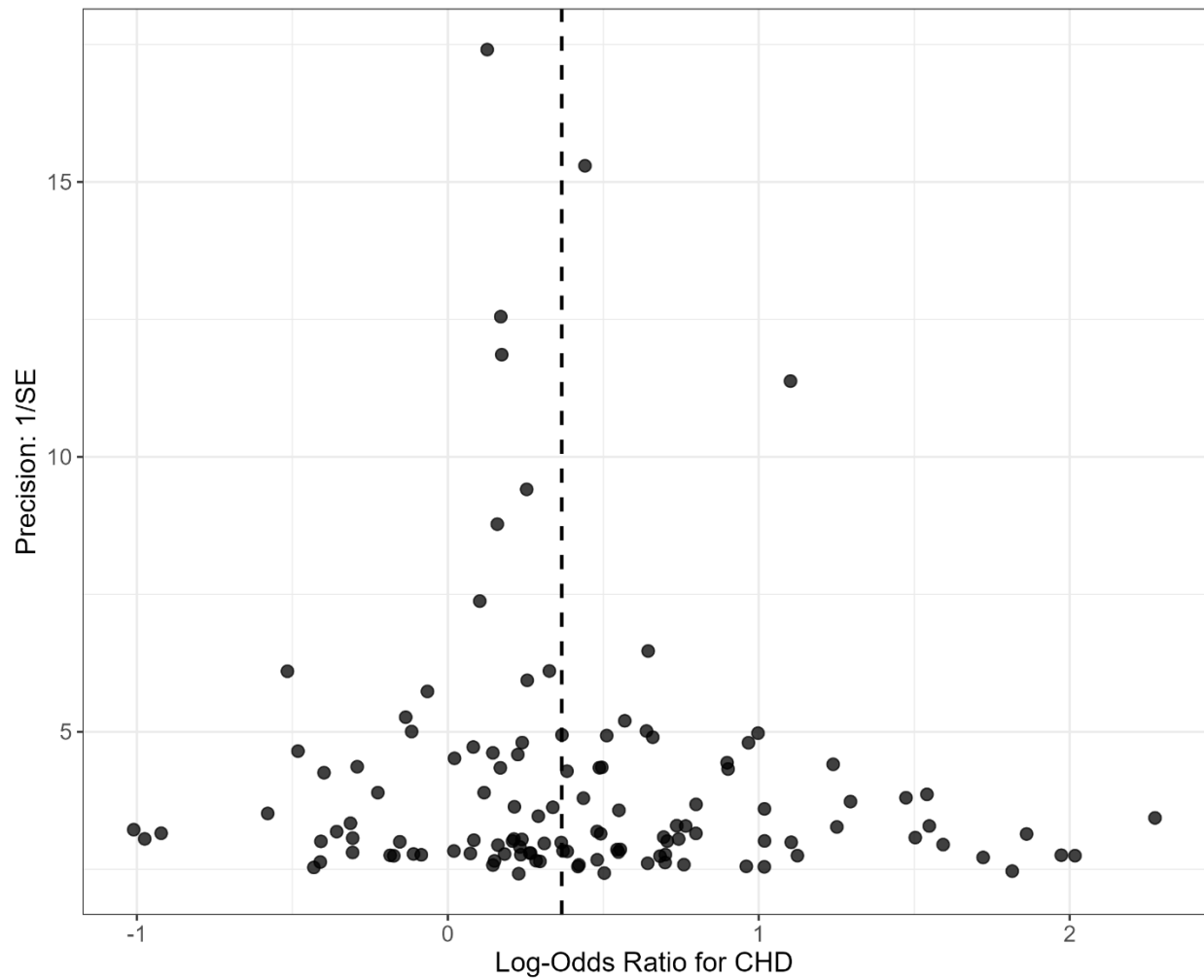
Supplementary Figure 1. Flow chart of sample selection of LDL-TG NMR data



Supplementary Figure 2. Two-sample MR leave-one-out sensitivity analysis for elevated LDL-TG to risk of CHD

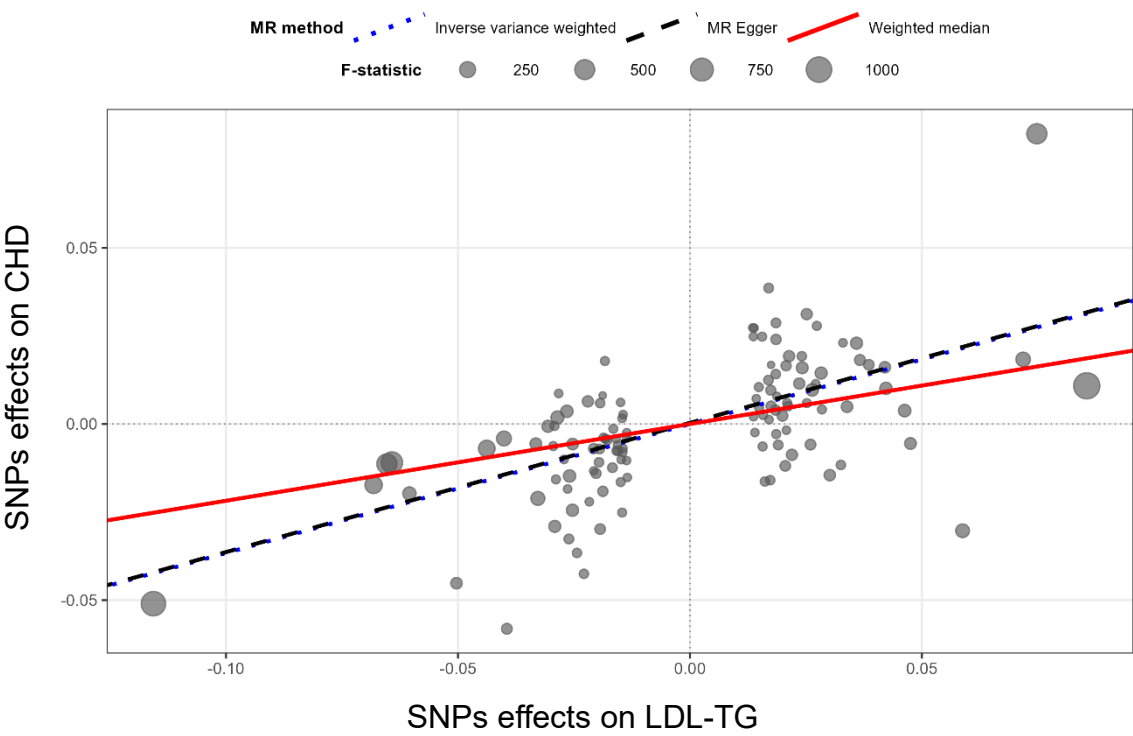


Supplementary Figure 3. Funnel plot to assess the robustness of using single SNPs as instrument variable in inverse variance weighted method



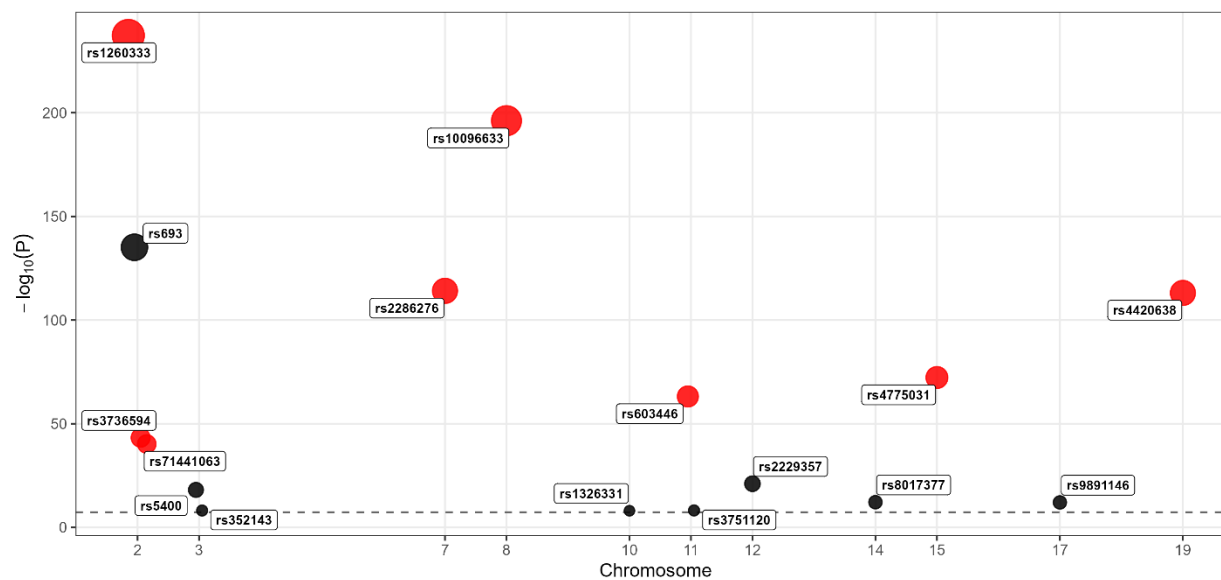
Notes: SE: standard error of the beta coefficient/effect estimate, CHD: coronary heart disease.

Supplementary Figure 4. MR scatter plots for the direction of elevated LDL-TG level to risk of CHD





Supplementary Figure 5. Comparison of F-statistics between coding-consequence (all SNPs) and non-coding sequence (selected SNPs)



Notes: black solid oval: SNPs in coding sequence, red solid oval: selected SNPs in non-coding sequence.

Supplementary Table 1. Example of cross validated SNPs in the coding sequence

| SNP       | Mapped Gene/Region | Reported trait                      | PubMed ID          | Study type        |
|-----------|--------------------|-------------------------------------|--------------------|-------------------|
| rs693     | APOB               | ApoB, TG, TC, LDL-C and lower HDL-C | 28874158           | Meta-analysis     |
| rs5400    | SLC2A2             | Glycemic traits, metformin response | 27500523           | GWAS              |
| rs352143  | TWF2 / TLR9 region | Tuberculosis                        | 25616954           | Multicenter study |
| rs1326331 | EXOC6 / SEC15L1    | Late-onset Alzheimer disease        | 16385451           | Comparative study |
| rs3751120 | FLRT1, MACROD1     | CAD among African Americans         | 28396569           | Multicenter study |
| rs2229357 | INHBC, INHBE       | CAD risk general                    | 28714975           | GWAS              |
| rs8017377 | NYNRIN             | LDL-C                               | 20686565, 24097068 | GWAS              |
| rs9891146 | C17orf58           | Type 2 diabète susceptible variants | 29632382           | Meta-analysis     |

Supplementary Table 2. Example of cross validated SNPs in the non-coding sequence

| SNP                      | Mapped Gene   | Location   | Closest SNP              | Reported trait | PubMed ID | Sample size |
|--------------------------|---------------|------------|--------------------------|----------------|-----------|-------------|
| rs11631178,<br>rs8035006 | ALDH1A2; LIPC | chr15q21.3 | rs1800588,<br>rs10468017 | LDL-TG to CHD  | 30685440  | N=487       |

Supplementary Table 3. Full list of identified SNPs with mapped region

| SNP         | F-statistics | Mapped Gene/Region | In non-coding sequence | Chromosome |
|-------------|--------------|--------------------|------------------------|------------|
| rs4970829   | 59.86        | SARS1              | Yes                    | 1          |
| rs6587971   | 58.77        | Not mapped         | Yes                    | 1          |
| rs2061154   | 32.49        | LYPLAL1-AS1        | Yes                    | 1          |
| rs1260333   | 1083.68      | GCKR               | Yes                    | 2          |
| rs693       | 613.79       | APOB               | No                     | 2          |
| rs3736594   | 192.1        | MRPL33             | Yes                    | 2          |
| rs71441063  | 178.17       | ATRAID; CAD        | Yes                    | 2          |
| rs10195252  | 154.41       | COBLL1             | Yes                    | 2          |
| rs7081      | 132.29       | SLC5A6             | Yes                    | 2          |
| rs312965    | 77.24        | TDRD15             | Yes                    | 2          |
| rs111548358 | 72.81        | Not mapped         | Yes                    | 2          |
| rs10495713  | 67.87        | Not mapped         | Yes                    | 2          |
| rs66489250  | 63.76        | BABAM2             | Yes                    | 2          |
| rs10194882  | 55.3         | Not mapped         | Yes                    | 2          |
| rs17759655  | 54.98        | BABAM2             | Yes                    | 2          |
| rs55669835  | 40.68        | FAM117B            | Yes                    | 2          |
| rs35932591  | 37.06        | Not mapped         | Yes                    | 2          |
| rs55701660  | 31.92        | LINC01822          | Yes                    | 2          |
| rs6710941   | 31.85        | CENPA              | Yes                    | 2          |
| rs115111187 | 28           | TTLL4              | Yes                    | 2          |
| rs3732158   | 27.89        | HS1BP3             | Yes                    | 2          |
| rs16824590  | 27.39        | NOP58; SNORD70     | Yes                    | 2          |
| rs5400      | 77.03        | SLC2A2             | No                     | 3          |
| rs9845457   | 50.04        | MSL2               | Yes                    | 3          |
| rs13326165  | 32.37        | STAB1              | Yes                    | 3          |
| rs352143    | 32.08        | TWF2               | No                     | 3          |
| rs4683702   | 27.77        | Not mapped         | Yes                    | 3          |
| rs442177    | 118.14       | AFF1               | Yes                    | 4          |
| rs7694379   | 27.51        | Not mapped         | Yes                    | 4          |
| rs4450871   | 27.48        | Not mapped         | Yes                    | 4          |
| rs41275715  | 27.1         | MTTP               | Yes                    | 4          |
| rs3846662   | 49.86        | CERT1; HMGCR       | Yes                    | 5          |
| rs11134551  | 37.17        | Not mapped         | Yes                    | 5          |
| rs57079115  | 27.84        | Not mapped         | Yes                    | 5          |
| rs9271588   | 104.6        | HLA-DQA1           | Yes                    | 6          |
| rs2523589   | 96.22        | HLA-B; MIR6891     | Yes                    | 6          |
| rs1052248   | 91.48        | LST1; NCR3         | Yes                    | 6          |
| rs3094228   | 90.71        | HCP5               | Yes                    | 6          |
| rs628751    | 86.07        | Not mapped         | Yes                    | 6          |
| rs35372932  | 82.41        | HLA-DRB1           | Yes                    | 6          |

|            |        |                     |     |    |
|------------|--------|---------------------|-----|----|
| rs998203   | 68.24  | CHP1P2; IGF2R       | Yes | 6  |
| rs3134931  | 54.25  | NOTCH4              | Yes | 6  |
| rs2517600  | 45.51  | TRIM31; TRIM31-AS1  | Yes | 6  |
| rs2844795  | 41.64  | TRIM31; TRIM31-AS1  | Yes | 6  |
| rs885912   | 40.99  | TRIM26              | Yes | 6  |
| rs1233478  | 37.14  | LINC02829           | Yes | 6  |
| rs3749946  | 37.12  | HCP5; MICB-DT       | Yes | 6  |
| rs4895864  | 36.29  | L3MBTL3             | Yes | 6  |
| rs241429   | 31.68  | PSMB8; TAP2         | Yes | 6  |
| rs2072633  | 31.67  | CFB; CYP21A2; NELFE | Yes | 6  |
| rs1490384  | 28.19  | Not mapped          | Yes | 6  |
| rs10484552 | 27.73  | LINC02569           | Yes | 6  |
| rs9263717  | 27.49  | PSORS1C1            | Yes | 6  |
| rs6907751  | 27.38  | PEX6                | Yes | 6  |
| rs4895815  | 27.19  | Not mapped          | Yes | 6  |
| rs2286276  | 517.23 | TBL2                | Yes | 7  |
| rs10951261 | 50.36  | ADAP1; COX19        | Yes | 7  |
| rs6977319  | 32.46  | BAZ1B               | Yes | 7  |
| rs10229375 | 31.83  | MIR4284; STX1A      | Yes | 7  |
| rs13306387 | 28.16  | GCK; MYL7           | Yes | 7  |
| rs58053084 | 27.61  | GCK                 | Yes | 7  |
| rs12702029 | 27.44  | CHLSN; GPER1        | Yes | 7  |
| rs10096633 | 894.51 | Not mapped          | Yes | 8  |
| rs13273592 | 68.46  | Not mapped          | Yes | 8  |
| rs6987457  | 45.38  | Not mapped          | Yes | 8  |
| rs34370554 | 40.99  | Not mapped          | Yes | 8  |
| rs10087499 | 31.86  | Not mapped          | Yes | 8  |
| rs17729317 | 28.37  | TRPS1               | Yes | 8  |
| rs2782981  | 36.3   | Not mapped          | Yes | 10 |
| rs1326331  | 31.6   | EXOC6               | No  | 10 |
| rs1937917  | 27.91  | AKR1C5P             | Yes | 10 |
| rs603446   | 283.27 | ZPR1                | Yes | 11 |
| rs1784042  | 99.66  | SIDT2; TAGLN        | Yes | 11 |
| rs7950381  | 91.24  | SIK3                | Yes | 11 |
| rs174601   | 68.13  | FADS2               | Yes | 11 |
| rs34841529 | 37     | Not mapped          | Yes | 11 |
| rs3751120  | 32.25  | FLRT1; MACROD1      | No  | 11 |
| rs2229357  | 90.56  | INHBC; INHBE        | No  | 12 |
| rs4363657  | 73.49  | SLCO1B1             | Yes | 12 |
| rs12829378 | 36.31  | RFLNA               | Yes | 12 |
| rs11054397 | 31.7   | Not mapped          | Yes | 12 |
| rs11172156 | 27.62  | R3HDM2              | Yes | 12 |
| rs12298717 | 27.35  | RFX4                | Yes | 12 |

|            |        |                                      |     |    |
|------------|--------|--------------------------------------|-----|----|
| rs4942486  | 45.44  | BRCA2                                | Yes | 13 |
| rs12869880 | 32.14  | GAS6; GAS6-AS1                       | Yes | 13 |
| rs8017377  | 50.15  | NYNRIN                               | No  | 14 |
| rs4775031  | 325.18 | ALDH1A2                              | Yes | 15 |
| rs62020698 | 104.94 | FDPSP4; UBR1                         | Yes | 15 |
| rs11631178 | 100.64 | ALDH1A2; LIPC                        | Yes | 15 |
| rs55995747 | 81.86  | ALDH1A2; LIPC; LIPC-AS1              | Yes | 15 |
| rs12438071 | 76.84  | ALDH1A2; LIPC; LIPC-AS1              | Yes | 15 |
| rs2242069  | 58.78  | TP53BP1                              | Yes | 15 |
| rs2652836  | 37.22  | Not mapped                           | Yes | 15 |
| rs12594625 | 36.07  | ALDH1A2; AQP9; MTCYBP23;<br>MTND5P32 | Yes | 15 |
| rs12439377 | 27.14  | HAUS2; MYL12BP1                      | Yes | 15 |
| rs8035006  | 27.11  | LIPC                                 | Yes | 15 |
| rs6416553  | 23.84  | CRTC3                                | Yes | 15 |
| rs28629271 | 68.02  | PKD1L3                               | Yes | 16 |
| rs7188811  | 32.45  | C16orf96; UBALD1                     | Yes | 16 |
| rs9923575  | 32.41  | Not mapped                           | Yes | 16 |
| rs3759979  | 32.26  | COG8; NIP7; TMED6                    | Yes | 16 |
| rs255049   | 31.93  | DPEP3                                | Yes | 16 |
| rs1364063  | 31.88  | Not mapped                           | Yes | 16 |
| rs9891146  | 49.68  | C17orf58                             | No  | 17 |
| rs12937634 | 41.28  | ZNF652; ZNF652-AS1                   | Yes | 17 |
| rs17824149 | 32.61  | TM4SF5                               | Yes | 17 |
| rs55662831 | 28.24  | DNAH2                                | Yes | 17 |
| rs11867485 | 27.22  | CD300LG                              | Yes | 17 |
| rs62074057 | 23.88  | TBX21                                | Yes | 17 |
| rs4420638  | 512.62 | APOC1                                | Yes | 19 |
| rs516246   | 99.82  | FUT2                                 | Yes | 19 |
| rs11881756 | 41.66  | CEACAM16-AS1                         | Yes | 19 |
| rs4805885  | 40.52  | PEPD                                 | Yes | 19 |
| rs2967777  | 36.43  | MYO1F                                | Yes | 19 |
| rs7188     | 31.8   | KANK2                                | Yes | 19 |
| rs1529729  | 27.34  | SMARCA4                              | Yes | 19 |
| rs77617917 | 146.09 | PCIF1                                | Yes | 20 |
| rs680379   | 36.78  | LINC01723                            | Yes | 20 |
| rs2143877  | 28.26  | Not mapped                           | Yes | 20 |
| rs73122720 | 23.92  | RNU6ATAC38P; WFDC3                   | Yes | 20 |
| rs5760176  | 27.47  | GSTT4; GSTTP2                        | Yes | 22 |

Notes: yellow rows are all SNPs that are in the coding sequence, green rows are selected SNPs that are in the non-coding sequence.

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